The Width Barrier

A Conditional Decidability Result for $\mathcal{ALCI}_{\mathrm{RCC5}}$, and the Unity of Its Two Open Directions

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Abstract

Concept satisfiability for $\mathcal{ALCI}_{\mathrm{RCC5}}$ — the description logic $\mathcal{ALCI}$ over the five RCC5 base relations under abstract composition-table semantics — has been open since Wessel (2002/2003). This report records the current, honestly-delimited state of one line of attack. We establish a *conditional* decidability theorem: $\mathcal{ALCI}_{\mathrm{RCC5}}$ satisfiability is decidable provided a single property holds — *bounded live width* (the obligation we call F6) — and we note that the soundness half of the underlying reduction is machine-checked in Lean 4 with no unproved steps. We then analyse F6 itself. Two facts about the RCC5 composition table, both exhaustively verified, drive the analysis: (i) the only compositions that *determine* a relation (are singletons) all involve a vertical (PP/PPI) argument, and PO is never a determined value; (ii) consequently the "determined" structure a presentation may carry for free (*shadows*) is confined to the vertical axis, while the costly *live* structure lives on the horizontal (DR/PO) axis. Building on these we show that the width barrier *reduces to a definability question*: F6 fails if and only if some $\mathcal{ALCI}_{\mathrm{RCC5}}$ concept can *force* unboundedly large rigid horizontal configurations. The substrate for such configurations is free (every DR/PO network is realizable), and horizontal chains *block* exactly as vertical ones do; so the entire barrier concentrates into whether the logic can pin rigid horizontal coordinates. Finally we observe that this is precisely the obstruction on the *undecidability* side as well: the reason no grid-encoding transfers to the abstract semantics (Wessel 2002/2003; Lutz–Wolter 2006) is the same looseness. The decidability keystone and the undecidability obstruction are one fact seen from two sides. We do not resolve F6; we localize it, certify everything around it, and place it on the knife's edge between the two outcomes.

1 Introduction

The Region Connection Calculus fragment RCC5 partitions the relationships between two non-empty spatial regions into five jointly-exhaustive, pairwise-disjoint base relations. The description logic $\mathcal{ALCI}_{\mathrm{RCC5}}$ takes these five as its roles (inverse roles are absorbed, since each relation's converse is again a base relation) and interprets them over *abstract* models: structures in which every pair of distinct elements is labelled by exactly one base relation, subject only to the RCC5 composition table and converse laws. Whether concept satisfiability in this logic is decidable has been open for over two decades [1, 2]. No undecidability reduction is known to apply to the abstract semantics, and no decision procedure has been certified.

This report is not a claimed resolution. It is a *structural status report*: an account of where the difficulty now sits, written to be handed to someone new. Its contributions are three.

1. A **conditional decidability theorem** (§5): $\mathcal{ALCI}_{\mathrm{RCC5}}$ satisfiability is decidable if the *bounded live width* property F6 holds. The soundness direction of the reduction — that a finite, well-

formed, composition-closed, faithfully-labelled certificate yields a genuine RCC5 model of the input concept — is fully machine-checked in Lean 4.

2. A **reduction of F6 to a definability question** (§§6–7): using two exhaustively verified facts about the composition table, we show that F6 fails exactly when some concept can *force* unboundedly large rigid horizontal configurations, and that the combinatorial substrate for such configurations is otherwise free.

3. A **unification of the two open directions** (§8): the forcing question that F6 reduces to is the same obstruction that blocks every attempt to prove the problem *undecidable*. Bounding it one way yields decidability; exploiting it the other way would yield undecidability; and the reason neither has happened is a single fact about the composition table.

Throughout, we distinguish clearly between what is machine-checked, what is exhaustively verified by finite computation, and what is argued.

2 Preliminaries

2.1 RCC5 and its composition table

The five base relations are DR (discrete — disjoint interiors), PO (partial overlap — share a part, neither contained in the other), PP (proper part), PPI (inverse proper part), and EQ (equal). Converse is the involution $\overline{\mathsf{DR}} = \mathsf{DR}$, $\overline{\mathsf{PO}} = \mathsf{PO}$, $\overline{\mathsf{EQ}} = \mathsf{EQ}$, $\overline{\mathsf{PP}} = \mathsf{PPI}$, $\overline{\mathsf{PPI}} = \mathsf{PP}$. The composition $\mathrm{comp}(R, S)$ collects every base relation that can hold between x and z given $x\,R\,y$ and $y\,S\,z$; it is derivable from the subset semantics on non-empty regions and is displayed in Table 1.

comp	EQ	DR	PO	PP	PPI
EQ	{EQ}	{DR}	{PO}	{PP}	{PPI}
DR	{DR}	{EQ, DR, PO, PP, PPI}	{DR, PO, PP}	{DR, PO, PP}	{DR}
PO	{PO}	{DR, PO, PPI}	{EQ, DR, PO, PP, PPI}	{PO, PP}	{DR, PO, PPI}
PP	{PP}	{DR}	{DR, PO, PP}	{PP}	{EQ, DR, PO, PP, PPI}
PPI	{PPI}	{DR, PO, PPI}	{PO, PPI}	{EQ, PO, PP, PPI}	{PPI}

Table 1: The RCC5 composition table, $\mathrm{comp}(\mathrm{row}, \mathrm{col})$. Rows are the first argument, columns the second. Computed by exhaustive enumeration over finite region domains.

An *atomic RCC5 network* on a set V assigns to each ordered pair of distinct elements exactly one base relation $N(u, v)$, satisfying converse coherence $N(v, u) = \overline{N(u, v)}$ (R2) and composition closure $N(u, w) \in \mathrm{comp}(N(u, v), N(v, w))$ for all distinct u, v, w (R3); under *strong-EQ* semantics EQ occurs only on the diagonal (R1). We use the standard external input of the whole subject:

Proposition 1 (Patchwork property; Renz–Nebel)**.** *A path-consistent atomic RCC5 constraint network is globally realizable: it has a model.*

2.2 $\mathcal{ALCI}_{\mathrm{RCC5}}$ syntax and semantics

Concepts in negation normal form are generated by

$$C, D \ ::= \ \top \mid \bot \mid A \mid \neg A \mid C \sqcap D \mid C \sqcup D \mid \exists R.C \mid \forall R.C,$$

where A is a concept name and $R \in \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$ (the reflexive EQ is not used as a role between distinct elements). An interpretation $\mathcal{I} = (\Delta, \rho, \cdot^{\mathcal{I}})$ has a non-empty domain Δ, an atomic RCC5 network ρ on Δ, and concept-name extensions $A^{\mathcal{I}} \subseteq \Delta$. The role reading is $(\exists R.C)^{\mathcal{I}} = \{x \mid \exists y\,(\rho(x, y) = R \wedge y \in C^{\mathcal{I}})\}$ and dually for $\forall$; because ρ is atomic, the extension of role R is exactly the set of pairs labelled R. C_0 is *satisfiable* if $C_0^{\mathcal{I}} \neq \emptyset$ for some such $\mathcal{I}$. This is the abstract composition-table semantics of [1]; it is what makes the problem different from the topological logics of [2], whose undecidability does not transfer.

3 The split-forest presentation and the two axes

The line of attack organizes any model along two axes. The *vertical* axis is nesting: PP and PPI form a forest, since PP is a strict partial order ($\mathrm{comp}(\mathsf{PP}, \mathsf{PP}) = \{\mathsf{PP}\}$, so PP is transitive, and it is irreflexive under strong-EQ). The *horizontal* axis is DR and PO: the relations between elements that are not nested in one another. A model thus presents as a forest of nested regions with horizontal cross-links between incomparable ones; this is the *split-forest presentation.* The entire difficulty of the problem is captured by one asymmetry, which we now make precise.

4 Shadows and live width

The composition table forces some relations and leaves others free. The crucial instance is the pair

$$\mathrm{comp}(\mathsf{PP}, \mathsf{DR}) = \{\mathsf{DR}\}, \qquad \mathrm{comp}(\mathsf{PPI}, \mathsf{DR}) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PPI}\}. \tag{1}$$

The first is a singleton: if $u\ \mathsf{PP}\ w$ and $w\ \mathsf{DR}\ v$ then $u\ \mathsf{DR}\ v$ is *forced* — discreteness propagates *into parts.* The second is not: discreteness does not propagate into wholes. This one-directional determinism is the source of both the method's power and its difficulty.

Definition 2 (Shadow / determined edge)**.** In an atomic RCC5 network N on V, an edge $\{u, v\}$ is *determined* (a *shadow*) if its label is fixed by path-consistency from N restricted to $V \setminus \{u, v\}$ together with the two elements' other edges. A sufficient local condition is the existence of $w \in V$ with $|\mathrm{comp}(N(u, w), N(w, v))| = 1$. An edge that is not determined is *live.*

A shadow costs nothing in a finite presentation: it is recoverable and need not be independently stored or checked. Hence the size that matters for a blueprint is not a node's degree but its number of live neighbours.

Definition 3 (Live width; the property F6)**.** The *live width* of a split-forest presentation is the supremum, over its interface clusters, of the number of live occurrences that must coexist in one cluster. Property **F6 (Bounded Live Width)**: there is a computable K such that every satisfiable concept C_0 has a model whose split-forest presentation has live width at most $K(|C_0|)$.

5 The conditional decidability theorem

The reduction built over rounds of this project, and its soundness core now certified in Lean 4, yields the following. We state it at the level of certificates; the formal development is `formal/Round19Transport.lean` (roughly 4,500 lines, zero `sorry`, no added axioms).

Theorem 4 (Conditional decidability)**.** *If F6 holds, then $\mathcal{ALCI}_{\mathrm{RCC5}}$ concept satisfiability is decidable.*

Sketch, with the certified content marked. A *certificate* is a finite *catalogue* (core network, attachment templates with a gluing normal form, and a template-indexed steering function) together with a plan; unfolding a plan produces a split-forest presentation. The development establishes, *in the Lean kernel*:

1. *Generation by construction.* For a structurally sound catalogue and a valid plan, the unfolded certificate is well-formed and *faithful* (its pattern values agree with the recorded interface rows) — `build_wellformed`, `build_faithful`.
2. *Catalogue soundness.* A finite composition check on the catalogue alone (the *S-condition*, independent of the plan length) plus faithfulness implies the full per-unfolding closure condition — `scat_scond` — hence the unfolded frame is a proper atomic RCC5 network (composition-closed, converse-coherent, EQ-free): `frame_closed`, `catalogue_soundness`.
3. *Truth.* A Hintikka labelling of the frame with the target concept at a root yields a genuine RCC5 model of that concept — `truth_lemma`, `sat_from_hintikka`; and the induced frame is verified to satisfy R1–R3 (`certInterp_rcc5`).
4. *Faithful abstraction.* Satisfiability and Hintikka-realizability coincide (`satisfiable_iff_hintikkaP`): a model always induces a Hintikka labelling, so the certificate loses nothing.

The decision procedure enumerates finite catalogues, checks the finite S-condition and a Hintikka labelling, and accepts iff a valid one carries C_0 at a root. Soundness (accept $\Rightarrow$ satisfiable) is (1)–(3), certified. Completeness (satisfiable $\Rightarrow$ accept) needs every satisfiable C_0 to admit a *finite* catalogue; this is the obligation `CompletenessObligation` in the development, and it follows from F6 (which bounds the catalogue, giving a finite search space and termination) together with a minor uniformization property W2′ affecting only extraction faithfulness — a failure of which can only cause over-rejection, never unsoundness. □

5.1 Soundness and completeness: what is certified

It is worth separating the two directions of the reduction and stating exactly what the Lean development establishes for each, since the completeness side is more subtle than a single "machine-checked" would suggest.

Soundness — a valid finite certificate yields a genuine RCC5 model — is machine-checked in full (items (1)–(3) above: `catalogue_soundness`, `frame_closed`, `sat_from_hintikka`).

Completeness splits into two layers, and they have different status.

- **Abstraction completeness (infinitary) — certified.** Allowing an arbitrary, possibly infinite Hintikka labelling, completeness *is* machine-checked: `satisfiable_iff_hintikkaP` proves that C_0 is satisfiable *if and only if* some RCC5 frame carries a Hintikka labelling with C_0 at a point — both directions, no `sorry`. The non-trivial half, `model_hintikkaP` (every model induces a Hintikka labelling), is immediate once one observes that the Hintikka clauses are exactly the recursion clauses of satisfaction (§5, item (4)). Thus the type-system abstraction is *provably faithful*: it captures satisfiability exactly and loses nothing relative to models.
- **Finitary completeness — open, and equal to** $\mathrm{F6} \wedge \mathrm{W2}'$**.** Requiring a *finite* certificate, as decidability demands, completeness is precisely the obligation `CompletenessObligation`: every satisfiable C_0 admits a finite catalogue, plan, and Hintikka labelling with C_0 at a root. This is *stated* in the development but deliberately *not proved* and *not axiomatized*; it is the conjunction of F6 (bounded width, which makes the catalogue finite and the search terminating) with W2′ (uniformization, which makes finite extraction faithful, and whose failure can only over-reject).

Direction	Lean witness	Status
Soundness: certificate ⇒ model	`sat_from_hintikka`	certified
Completeness (abstraction): model ⇔ Hintikka	`satisfiable_iff_hintikkaP`	certified
Completeness (finitary): sat ⇒ finite certificate	`CompletenessObligation`	open = $\mathrm{F6} \wedge \mathrm{W2}'$

In one line: the reduction is *sound* and *complete modulo* $\mathrm{F6} \wedge \mathrm{W2}'$ — with soundness and the infinitary abstraction-completeness both certified, and the sole unformalized remainder being exactly $\mathrm{F6} \wedge \mathrm{W2}'$, which the rest of this report analyses.

Remark 5. Theorem 4 is a genuine reduction, not a restatement: F6 is a single, precisely-posed property, and everything around it is machine-checked. What is *not* established is F6 itself (with the minor W2′). The rest of this report analyses it.

6 The anatomy of forcing

We now isolate what the composition table can and cannot *force*. Call a composition $\mathrm{comp}(R, S)$ *determining* if it is a singleton.

Proposition 6 (Determination is vertical)**.** *Among the sixteen compositions of proper (non-*EQ*) relations, exactly four are determining:*

$$\mathrm{comp}(\mathsf{DR}, \mathsf{PPI}) = \{\mathsf{DR}\}, \quad \mathrm{comp}(\mathsf{PP}, \mathsf{DR}) = \{\mathsf{DR}\}, \quad \mathrm{comp}(\mathsf{PP}, \mathsf{PP}) = \{\mathsf{PP}\}, \quad \mathrm{comp}(\mathsf{PPI}, \mathsf{PPI}) = \{\mathsf{PPI}\}.$$

Every one of them has a vertical (PP *or* PPI*) argument. No composition with both arguments horizontal (*$\in \{\mathsf{DR}, \mathsf{PO}\}$*) is determining, and* PO *is never the value of a determining composition.*

Proof. Exhaustive inspection of Table 1; verified by finite computation. The horizontal block $\{\mathsf{DR}, \mathsf{PO}\} \times \{\mathsf{DR}, \mathsf{PO}\}$ gives $\mathrm{comp}(\mathsf{DR}, \mathsf{DR})$ and $\mathrm{comp}(\mathsf{PO}, \mathsf{PO})$ (all five relations) and $\mathrm{comp}(\mathsf{DR}, \mathsf{PO}) = \mathrm{comp}(\mathsf{PO}, \mathsf{DR})$'s companions of size three; none is a singleton. The four singletons are as listed, each with a PP/PPI leg. Scanning the table, no cell equals $\{\mathsf{PO}\}$. □

Proposition 6 explains a phenomenon observed experimentally: attempts to force a region to acquire unboundedly many neighbours succeed only in producing shadows. Any *forced* edge arises from a determining composition (or an intersection of constraints that pins a value), and by Proposition 6 the only fully determining compositions run through the vertical axis — so forced structure is shadow structure, confined to nesting. Concretely, a construction in which a distinguished "spy" region is forced discrete from every level of an ascending tower makes the spy's degree grow without bound, yet each such edge is a shadow (forced by $\mathrm{comp}(\mathsf{PP}, \mathsf{DR}) = \{\mathsf{DR}\}$ through the tower); the *live* width stays bounded.

There is, however, a subtlety that keeps the barrier open. Determination and *distinctness* are different.

Proposition 7 (Distinctness can be horizontal)**.** *Two same-typed occurrences are mergeable (identifiable, set to* EQ*) unless some constraint excludes* EQ *between them. A shared horizontal anchor excludes* EQ*: if* $u \,\mathsf{DR}\, w$ *and* $w \,\mathsf{PO}\, v$ *then* u, v *satisfy* $\mathrm{comp}(\mathsf{DR}, \mathsf{PO}) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}\}$*, which excludes* EQ *(forcing* $u \neq v$*) yet is not a singleton (the edge* $\{u, v\}$ *remains live).*

Proof. From Table 1, $\mathrm{comp}(\mathsf{DR}, \mathsf{PO}) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}\} \not\ni \mathsf{EQ}$ and $|\{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}\}| = 3$. □

Thus *determination* is strictly vertical (Prop. 6) but *distinctness* extends to the horizontal axis (Prop. 7). The two notions come apart, and it is precisely in the gap between them that F6 lives: a live horizontal crowd — many occurrences, pairwise distinct by horizontal anchors, with live (undetermined) mutual relations — is not ruled out by Proposition 6.

7 The width barrier reduces to horizontal forcing

We now follow the live-crowd possibility to its floor. Three facts, each finite-checkable, combine into a reduction.

Proposition 8 (The substrate is free)**.** *Every* $\{\mathsf{DR}, \mathsf{PO}\}$*-labelled network is path-consistent, hence realizable by Proposition 1. Moreover, for every* n *there is such a network on* n *nodes in which no two nodes are mergeable (all rows distinct): an irreducible live horizontal crowd of size* n*.*

Proof. For $R, S \in \{\mathsf{DR}, \mathsf{PO}\}$ one has $\{\mathsf{DR}, \mathsf{PO}\} \subseteq \mathrm{comp}(R, S)$ (rows DR, PO of Table 1), so any $\{\mathsf{DR}, \mathsf{PO}\}$-labelling meets every composition constraint: path-consistency is immediate. Realizability follows from Proposition 1. For irreducibility, a $\{\mathsf{DR}, \mathsf{PO}\}$ network is a graph (edge $= \mathsf{PO}$, non-edge $= \mathsf{DR}$); a graph with pairwise-distinct neighbourhood rows has no two mergeable nodes, and such graphs exist at every size. □

Proposition 9 (Horizontal chains block)**.** *The concept* $\exists \mathsf{PO}.\top \sqcap \forall \mathsf{PO}.\exists \mathsf{PO}.\top$*, which demands an endless horizontal successor chain, is satisfiable by a finite model. More generally, any forced* PO*-chain admits a finite looping model.*

Proof. $\mathrm{comp}(\mathsf{PO}, \mathsf{PO})$ is the whole algebra (Table 1), and in particular contains EQ; hence a PO-chain may close a loop ($w_i = w_j$ is composition-consistent). With finitely many concept types, any chain repeats a type, and a repeated type with mergeable surroundings closes the loop, exactly as in the vertical blocking (§3). A two- or three-element cyclic model satisfies the displayed concept. □

Propositions 8 and 9 isolate the issue. A large live horizontal crowd is trivial to *exhibit* (Prop. 8) but cannot be produced merely by *demanding* more occurrences, because horizontal chains fold back on themselves (Prop. 9). To prevent the fold one would have to make the crowd members pairwise *non-mergeable in every model* — to give each a permanent address, a rigid horizontal coordinate. By Proposition 6 the composition table offers no lever to pin a horizontal coordinate: it cannot determine a horizontal relation, and PO is never forced. We arrive at the central structural statement.

Observation 10 (The width barrier is a forcing question)**.** Bounded live width (F6) fails if and only if some $\mathcal{ALCI}_{\mathrm{RCC5}}$ concept *forces* arbitrarily large *rigid* horizontal configurations — unboundedly many occurrences that must coexist at one interface and that no model may merge. Equivalently:

$$\textit{F6 counterexample} \iff \textit{a concept forcing an unbounded rigid } \{\mathsf{DR}, \mathsf{PO}\} \textit{ graph},$$
$$\textit{F6 proof} \iff \textit{no concept forces such a graph.}$$

The combinatorial substrate is free (Prop. 8); the mere generation of occurrences blocks (Prop. 9); so the whole barrier concentrates into whether the logic can force rigid horizontal coordination, a *definability* question about the expressive power of $\mathcal{ALCI}_{\mathrm{RCC5}}$.

Observation 10 is a reduction, not a theorem in the formal sense: the forward direction (unbounded width $\Rightarrow$ forced irreducible crowds) is the definition of live width unwound; the content is that the converse obstruction is a pure forcing question, because everything *except* forcing —

realizability and non-repetition — has been discharged by Propositions 8 and 9. We state it as an observation to mark that "forces" is used in its intuitive model-theoretic sense; making it a formal equivalence is itself part of the open problem.

8 One fact, two faces

The forcing question of Observation 10 is not new to the subject. It is the exact obstruction on the *undecidability* side.

To prove a logic undecidable one typically forces it to encode an unbounded, rigidly-coordinated grid on which a Turing computation (or a tiling) is written. For RCC8 with *topological* semantics this succeeds [2]. For the *abstract* composition-table semantics of $\mathcal{ALCI}_{\mathrm{RCC5}}$ (and $\mathcal{ALCI}_{\mathrm{RCC8}}$) it does not, and the reason was identified by [1]: the composition table cannot force the *coincidence condition* — that "east-of-north" and "north-of-east" name the same cell — because it cannot pin the relevant relations to single values. That is rigid horizontal coordination, and its unavailability is exactly Proposition 6: no horizontal determination, PO never forced.

Observation 11 (The knife's edge)**.** The bounded-width property F6 (decidability side) and the non-forcibility of a coordinate grid (undecidability side) are the same fact about $\mathcal{ALCI}_{\mathrm{RCC5}}$: *the abstract composition table is too loose to pin rigid horizontal structure.* If this looseness can be turned into a bound, one obtains decidability (Theorem 4); if it could be circumvented and a grid forced, one would obtain undecidability. The two open directions are one question seen from two sides, and the reason neither has been settled in two decades is that the same fact must be turned one way or the other.

9 Discussion

Why the combinatorics did not fit. Attempts to attack the width bound with Ramsey-type combinatorics were unsuccessful, and Observation 10 explains why by type mismatch. Ramsey theory concerns structure that is *unavoidable* in a large fixed object; F6 concerns structure that a formula can *force*. These differ in quantifier order: "every large graph contains X" versus "some formula makes every model contain X." The tools that fit the latter are model-theoretic: locality (Gaifman/Hanf), Ehrenfeucht–Fraïssé games, the composition method, and the finite/bounded-model-property machinery — together with the grid-encodability analysis in which [1, 2] already sit. This is a different toolbox from the model-*construction* techniques used to build the certificate layer, and it is the one Observation 11 indicates. We do not claim it succeeds; we claim it is the correct room to enter.

What is and is not established. Certified (Lean, no gaps): the soundness of the certificate-to-model pipeline and the faithfulness of the Hintikka abstraction (§5). Exhaustively verified (finite computation): Table 1 and Propositions 6, 7, 8, 9. Argued (structural, intuitive "forces"): Observations 10 and 11. Open: F6 itself, hence decidability. The honest one-line status is unchanged from the project's standing label: *strongly supported, not certified.*

Value independent of F6. Theorem 4 is a conditional result that stands regardless of F6's fate, with its soundness half machine-certified. Observation 11 is, to our knowledge, a new unification: it identifies the decidability keystone and the undecidability obstruction as one fact. Together they turn a formless two-decade question into a single, well-posed, two-sided lemma with a certified surrounding and a named toolbox. If the problem is ever settled we expect the resolution to begin from a map of this shape.

10 Conclusion

$\mathcal{ALCI}_{\mathrm{RCC5}}$ satisfiability is decidable if the live width of its models is bounded (Theorem 4), and the soundness of that reduction is machine-checked. Whether the width is bounded reduces to whether the logic can force rigid horizontal coordination (Observation 10), which is exactly the obstruction that blocks proving the problem undecidable (Observation 11). The composition table's inability to pin rigid horizontal structure is the single fact underneath both open directions. We have not turned that fact into a bound; we have shown that doing so, in either direction, is the whole of what remains.

References

[1] M. Wessel. *On spatial reasoning with description logics — position paper* and technical report (report7), 2002/2003. Introduces $\mathcal{ALCI}_{\mathrm{RCC}}$ under abstract composition-table semantics and identifies the coincidence-condition obstruction to grid encoding.

[2] C. Lutz and F. Wolter. *Modal Logics of Topological Relations.* Logical Methods in Computer Science, 2006. Undecidability for topological RCC logics; the reductions do not transfer to the abstract semantics.

[3] J. Renz and B. Nebel. *On the complexity of qualitative spatial reasoning: a maximal tractable fragment of the region connection calculus.* Artificial Intelligence, 1999. Source of the patchwork property.

Companion artifacts: the machine-checked development `formal/Round19Transport.lean`; the width-attack probes `verification/python/wp35_f6_width_attack.py`, `wp36_determination_vs_distinctness.py`, `wp37_f6_reduces_to_forcing.py`; and the plain-language companion `WHY_ITS_HARD.md`. This report is deliberately at the boundary of formal and expository; where it and the machine-checked artifacts disagree, the artifacts win.